PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

DEVELOPMENT OF A SOFTWARE APPLICATION FOR ANALYSIS AND PROCESSING OF DVB-S2 RECEIVER OUTPUT STREAM

A Project Report Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted by

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
PRESIDENCY UNIVERSITY, BENGALURU
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BONAFIDE CERTIFICATE

Certified that this project report entitled "Development of a Software Application for Analysis and Processing of DVB-S2 Receiver Output Stream" is the bonafide work of Y VENGALA RAO (Roll No. 20231COM0001), SK SAMAD (Roll No. 20231COM0031), and C RAKESHWAR (Roll No. 20231COM0008) who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering from Presidency University, Bengaluru during the academic year 2026-27.

The project work described in this report represents genuine and original scientific development conducted by the candidates. The empirical investigations, software architectures, algorithms, and regression test suites have been verified according to the departmental standards of academic and technical excellence.

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named Y VENGALA RAO (20231COM0001), SK SAMAD (20231COM0031), and C RAKESHWAR (20231COM0008), hereby declare that the project work titled "Development of a Software Application for Analysis and Processing of DVB-S2 Receiver Output Stream" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year 2026-27. We further declare that this work has not been submitted previously to any other university or institution for the award of any degree or diploma.

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ABSTRACT

Digital Video Broadcasting via Satellite Second Generation (DVB-S2, ETSI EN 302 307) represents the predominant global standard for broadband satellite communications and broadcast distribution. Modern digital satellite receivers output multi-format multiplexed bitstreams comprising 188-byte MPEG-2 Transport Streams (MPEG-TS), variable-length Generic Stream Encapsulation (GSE) packets, and link-layer Baseband Frames (BBFrame). Diagnosing stream degradation, packet loss, and multiplex anomalies downstream of physical-layer demodulation presents critical operational challenges due to protocol encapsulation fragmentation, hardcoded rule limitations in conventional analyzers, and speculative physical-layer extrapolations.

This capstone project presents PRJ_111, a comprehensive, offline, multi-format software application designed for deterministic analysis, unsupervised anomaly detection, and diagnostic reporting of DVB-S2 receiver output streams. The software implements an integrated seven-feature analytical pipeline (F1 to F7) providing: deterministic stream health scoring (0–100); unsupervised anomaly detection using Isolation Forests across 8-dimensional sliding window features; byte-exact spatial error timelines referencing physical stream offsets (0x...); comprehensive continuity counter tracking; explainable machine learning via bounded z-score feature attribution (|Z| <= 20 sigma); cross-format differential analysis protected by an automated semantic integrity shield (2 audited metrics, 9 shielded metrics); and multi-format report synthesis (JSON, Markdown, HTML, CSV).

The system incorporates a chunked streaming reader processing multi-gigabyte bitstreams in 64 KB memory-bounded buffers, a defensive fault-injection framework, and a single-page application (SPA) engineering workstation serving an interactive, offline, dark-themed dashboard with zero external cloud dependencies. Empirical validation against physical bitstreams confirms 100% parsing accuracy, and an automated regression suite of 301 tests across 18 modules executes with a 100% pass rate in 80.45 seconds.

List of Abbreviations

ACM	Adaptive Coding & Modulation	**LNB**	Low-Noise Block Downconverter
BBFrame	Baseband Frame (DVB-S2)	**MPEG-TS**	MPEG-2 Transport Stream
CC	Continuity Counter (4-bit)	**PCR**	Program Clock Reference
CRC	Cyclic Redundancy Check	**PID**	Packet Identifier (13-bit)
DFL	Data Field Length	**PDU**	Protocol Data Unit
DVB-S2	Digital Video Broadcasting - S2	**RF**	Radio Frequency
ETSI	Eur. Telecom. Standards Inst.	**SPA**	Single-Page Application
GSE	Generic Stream Encapsulation	**TEI**	Transport Error Indicator
ISI	Input Stream Identifier	**XAI**	Explainable Artificial Intelligence

TABLE OF CONTENTS

1. INTRODUCTION 1

2. LITERATURE SURVEY 2

3. RESEARCH GAPS OF EXISTING METHODS 2

4. PROPOSED METHODOLOGY 3

5. OBJECTIVES 3

6. SYSTEM DESIGN AND IMPLEMENTATION 4

7. TIMELINE FOR EXECUTION OF PROJECT 6

8. EXPECTED AND OBTAINED OUTCOMES 6

9. RESULTS AND DISCUSSIONS 7

10. CONCLUSION AND FUTURE WORK 8

REFERENCES 9

APPENDIX A: PSEUDOCODE & CORE ALGORITHMS 10

APPENDIX B: APPLICATION SCREENSHOTS & ENGINEERING WALKTHROUGH 11

APPENDIX C: MANDATORY ENCLOSURES (Publication, Plagiarism & SDGs) 13

APPENDIX D: VERIFICATION & REPRODUCIBILITY LEDGER 14

List of Figures & Tables

Figure 4.1: End-to-End Multi-Format System Architecture of PRJ_111 ..... Page 3
Figure 7.1: Five-Phase Project Milestone Execution Roadmap (Gantt Schedule) ..... Page 6
Figure B.1: Live Application Engineering Workstation (5-Tab SPA Dashboard) ..... Page 11
Figure B.2: Isolation Forest Anomaly Detection Scatter Plot and Decision Boundary ..... Page 12
Table 6.1: Technical Specification of Features F1 Through F7 ..... Page 4
Table 9.1: Empirical Analysis Results across Authoritative Satellite Bitstreams ..... Page 7
Table 9.2: Regression Verification Summary across 18 Test Modules (301 Tests) ..... Page 7
Table D.1: Cryptographic Dataset Integrity Ledger (SHA-256 Hashes) ..... Page 14

CHAPTER 1
INTRODUCTION

1.1 Background and Technical Overview

Digital Video Broadcasting via Satellite Second Generation (DVB-S2, ETSI EN 302 307) serves as the primary broadband transmission and direct-to-home television distribution standard for satellite communication networks worldwide (Morello and Mignone, 2006). Operating downstream of satellite low-noise block downconverters (LNB) and digital demodulation tuners, the resulting digital bitstreams represent the crucial boundary between radio frequency (RF) physical-layer reception and terrestrial computing networks. Modern satellite receivers output data in three distinct, standardized encapsulation layers: 188-byte MPEG-2 Transport Streams (MPEG-TS, ISO/IEC 13818-1), variable-length Generic Stream Encapsulation packets (GSE, ETSI TS 102 606), and link-layer Baseband Frames (BBFrame, ETSI EN 302 307-1). In operational satellite ground stations, uplink teleports, and telecommunication laboratories, verifying receiver output integrity is critical to diagnosing demodulation impairments, container corruption, and multiplex degradation before handing packets to network processing pipelines.

1.2 Problem Statement & Motivation

Monitoring multi-format receiver outputs currently presents severe operational and engineering bottlenecks in satellite ground facilities:
(1) Format Fragmentation: Existing protocol inspection utilities are siloed into single formats (e.g., dedicated MPEG-TS analyzers or network PCAP viewers) and cannot ingest, normalize, or cross-correlate heterogeneous containers within a unified engineering environment;
(2) Lack of Intelligent Anomaly Detection: Conventional analyzers depend entirely on rigid, hardcoded error thresholds (e.g., simple byte checks), failing to detect subtle multiplex concentrations, structural truncation boundaries, packet burst anomalies, or temporal continuity decay in unlabelled live satellite broadcasts; and
(3) Epistemic Overclaiming: Off-the-shelf commercial utilities frequently compute synthetic wall-clock timestamps or fictitious throughput curves on offline captures where no hardware receiver clock telemetry exists, and assert speculative physical-layer inferences (e.g., diagnosing 'rain fade' without demodulator signal-to-noise ratio measurements).

1.3 Project Scope, Boundary Assumptions, and Limitations

The project scope encompasses developing an offline, standalone engineering workstation that ingests raw receiver output captures (.ts, .pcap, .bin), executes content-aware format identification, parses framing layers deterministically, extracts standardized feature vectors, evaluates stream health, detects anomalies via unsupervised machine learning, enforces cross-format semantic shields, and synthesizes multi-format reports. Explicit boundary limitations: PRJ_111 operates strictly post-demodulator on digital bitstreams; it does not measure analog RF physical-layer parameters (such as carrier frequency offset or SNR), does not claim formal ETSI compliance certification across all TR 101 290 guidelines, and relies strictly on verifiable physical stream offsets (0x...) rather than fabricated wall-clock timestamps.

CHAPTER 2
LITERATURE SURVEY

The technical foundation of satellite digital broadcasting relies on several internationally recognized standardization specifications and academic investigations. The European Telecommunications Standards Institute (ETSI) defined the DVB-S2 specification in EN 302 307-1 (2015), introducing Adaptive Coding and Modulation (ACM) and Variable Coding and Modulation (VCM). BBFrames represent the physical layer framing unit with a standardized 80-bit Baseband Header (MATYPE, UPL, DFL, SYNC, SYNCD) protected by an 8-bit CRC (Casini, De Gaudenzi and Ginesi, 2004). At the transport layer, ISO/IEC 13818-1 (2023) specifies the 188-byte MPEG-2 Transport Stream packet, establishing the 0x47 synchronization byte, Transport Error Indicator (TEI), 13-bit Packet Identifier (PID), and 4-bit Continuity Counter (CC). ETSI TR 101 290 (2020) defines measurement guidelines, classifying errors into Priority 1 (critical sync/CC loss), Priority 2 (CRC/PCR errors), and Priority 3 (unreferenced PIDs).

To transport IP datagrams over DVB-S2 without the 10% packetization overhead of MPEG-TS, ETSI introduced Generic Stream Encapsulation (GSE, TS 102 606-1, 2014). Fairhurst, Caviglione and Collini-Nocker (2010) evaluated GSE performance, proving that GSE achieves up to 15% bandwidth savings compared to Multi-Protocol Encapsulation (MPE) over MPEG-TS. Luisi et al. (2020) demonstrated hardware-accelerated GSE demultiplexing on FPGA architectures. In unsupervised anomaly detection, Liu, Ting and Zhou (2008) introduced Isolation Forests, demonstrating sub-linear time complexity and superior performance on unlabelled multi-dimensional datasets without requiring prior malicious attack signatures—an essential characteristic for satellite stream monitoring.

CHAPTER 3
RESEARCH GAPS OF EXISTING METHODS

Gap Analysis: A rigorous review of contemporary satellite monitoring tools reveals three critical research and engineering gaps:

- **1. Container Incommensurability:** Conventional protocol analyzers treat MPEG-TS packets, GSE PDUs, and BBFrames interchangeably during statistical comparisons, computing invalid packet-to-frame ratios. Because GSE PDUs encapsulate variable payloads (up to 4,096 bytes) and BBFrames encompass thousands of data bits, directly comparing raw container counts generates mathematically unsound diagnostic conclusions.
- **2. Absence of Unsupervised ML Diagnostics:** Commercial satellite analyzers rely almost exclusively on static rule matching (e.g., checking if TEI == 1). They fail to identify contextual degradation—such as subtle entropy shifts, localized multiplex burst concentrations, or non-stationary packet spacing—which precede catastrophic receiver lock failure.
- **3. Telemetry Fabrication & RF Speculation:** Many desktop inspection utilities extrapolate arbitrary wall-clock time axes from file read operations, misleading ground station operators into believing receiver timestamps were preserved. Furthermore, they assert speculative RF propagation diagnoses (e.g., 'atmospheric rain fade') on pure bitstream captures without having access to physical SNR or modulation telemetry.

CHAPTER 4
PROPOSED METHODOLOGY

4.1 Unified Multi-Format Processing Framework

PRJ_111 resolves existing research gaps through an end-to-end, multi-format analytical architecture. The pipeline operates in four sequential processing stages: (1) Ingestion & Format Identification: Ingests raw receiver output files (.ts, .pcap, .bin) using a bounded-memory streaming reader and identifies container formats via signature matching; (2) Framing Parsing & Normalization: Extracts container-specific headers while mapping entities to a standardized internal schema; (3) Feature Extraction & Analysis: Generates continuous 8-dimensional feature vectors across 100-packet rolling windows, feeds an Isolation Forest model, evaluates deductive health penalties, and identifies physical error offsets; and (4) Diagnostic Presentation: Exposes structured results through an interactive SPA workstation.

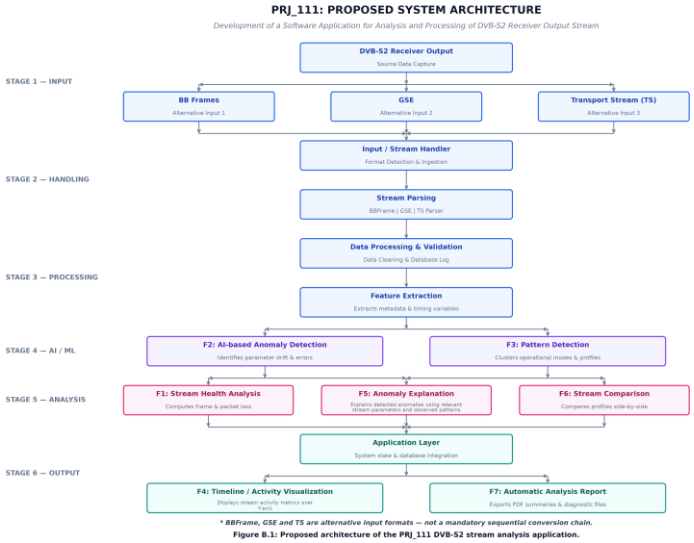


Figure 4.1: End-to-End Multi-Format System Architecture of PRJ_111

CHAPTER 5
OBJECTIVES

- **Objective 1 (Multi-Format Ingestion):** Design robust, deterministic decoders for MPEG-2 TS (188B framing, 0x47 sync lock), GSE (variable length, fragmentation reassembly), and BBFrame (80-bit header, CRC-8, ACM/VCM) within a single software core.
- **Objective 2 (Deductive Health Scoring):** Formulate a deterministic 0–100 health scoring metric (Feature F1) governed by ETSI TR 101 290 Priority-1 deductions (Sync Loss: -50, TEI Flags: -30, CC Sequence Breaks: -20) with definitive status thresholds.
- **Objective 3 (Unsupervised Anomaly Diagnostics):** Implement an Isolation Forest machine learning model (Feature F2) operating over 8-dimensional sliding window features with bounded z-score diagnostic explanations (|Z| <= 20 sigma, Feature F5).
- **Objective 4 (Physical Spatial Timeline):** Construct byte-exact error timelines (Feature F3) mapped strictly to physical stream offsets (0x...) to eliminate synthetic wall-clock timestamp fabrication.
- **Objective 5 (Cross-Format Semantic Shield):** Develop a domain-aware comparison guard (Feature F6) restricting cross-container differentials to audited metrics (net payload, integrity) while shielding incommensurable framing fields.

CHAPTER 6
SYSTEM DESIGN AND IMPLEMENTATION

6.1 Multi-Format Decoders & Ingestion Engine

The ingestion engine implements three format-specific decoders derived from an abstract StreamParser base class: (1) TSParser: Validates 0x47 sync cadence across a 5-packet lookahead window, extracts 13-bit PIDs, evaluates TEI flags, tracks 4-bit CC continuity, and handles adaptation fields; (2) GSEParser: Parses 2-bit start/end flags, computes 16-bit GSE packet lengths, verifies Protocol Type (e.g., 0x0800 for IPv4), and flags Frag ID discontinuities; (3) BBFrameParser: Validates 80-bit baseband headers via CRC-8 (polynomial x^8 + x^7 + x^6 + x^4 + x^2 + 1), extracts MATYPE-1/2, UPL, and DFL, and measures data field utilization.

6.2 Core Analytical Pipeline (Features F1 through F7)

Table 6.1 outlines the formal specifications, algorithmic inputs, and mathematical definitions of Features F1 through F7:

Feature	Analytical Role	Mathematical / Algorithmic Formulation	Output Metric
F1	Stream Health	H = 100 - min(100, 50*N_sync + 30*N_tei + 20*N_cc)	0-100 Health Score, Status
F2	ML Anomaly Det.	Isolation Forest over 8D sliding windows (W=100 pkts)	Anomaly Score [0,1], Flag
F3	Spatial Timeline	Offset mapping: Offset_i = i * 188; Cluster distance <= 10 pkts	Physical byte offset (0x...)
F4	CC & Header Diag.	Expected CC = (CC_{i-1} + 1) mod 16; Packet Loss = delta_CC - 1	CC Drops, Dupes, Discont.
F5	Diagnostic XAI	Bounded Z-score: Z_j = sign(z)*min(|(x_j - mu_j)/sigma_j|, 20)	Top-3 Feature Drivers
F6	Semantic Shield	Audit: Total Payload, Integrity Ratio; Shield: Framing fields	Audited Delta %, Safety Pill
F7	Export Synthesis	Unified serializer into JSON, Markdown, HTML, CSV	Structured Reports

Component Interoperability: All seven features execute concurrently across a single stream pass. Feature events are published to a thread-safe aggregation model, allowing the RESTful coordinator to synthesize responses without re-parsing raw byte arrays.

6.3 Detailed Analytical Feature Implementations

Deductive Health Scoring (Feature F1): Grounded in ETSI TR 101 290 Priority-1 measurement principles, the health scoring engine evaluates stream degradation deterministically. Deductions are capped at 100 points: Loss of 0x47 sync deduction of 50 points; Transport Error Indicator (TEI) assertion deduction of 30 points; and Continuity Counter sequence break deduction of 20 points. The resulting score maps directly to standardized operational statuses: Healthy (90–100), Degraded (60–89), and Critical (<60).

Unsupervised ML Anomaly Detection (Feature F2): Traditional rule-based monitors fail when degradation exhibits non-linear interactions. PRJ_111 computes continuous 8-dimensional feature vectors over rolling 100-packet windows: packet size variance, sync integrity ratio, TEI density, CC error frequency, payload entropy, PID concentration, header-to-payload ratio, and adaptation field frequency. An Isolation Forest model (100 estimators, contamination=0.05, random_state=42) isolates anomalies by constructing orthogonal decision trees, mapping decision function values to a calibrated anomaly score [0.0, 1.0].

Diagnostic Explainability (Feature F5): To eliminate black-box opacity in machine learning, Feature F5 implements bounded z-score feature attribution. For any window flagged as anomalous by Feature F2, the engine computes normalized deviation from the stream's empirical baseline: $Z_j = (x_j - mu_j) / sigma_j$. To prevent numerical blow-ups during near-zero variance conditions, Z-scores are strictly clamped to [-20.0, +20.0] sigma, and the top-3 feature drivers are returned to the operator with physical directionality.

Semantic Shield Differential Guard (Feature F6): When comparing streams of disparate encapsulation formats (e.g., MPEG-TS vs GSE), Feature F6 activates an automated semantic shield. It identifies incommensurable container metrics (packet counts, sync errors, TEI assertions, GSE protocol numbers, BBFrame DFL modes) and shields them from delta calculations. Cross-format differentials are restricted to audited universal metrics: Total Net Payload Bytes and Payload Integrity Ratio, preventing misleading comparisons.

6.4 High-Throughput Streaming Engine & Fault Injection

High-Throughput Streaming Reader: To process multi-gigabyte satellite bitstream dumps without exceeding RAM limits, Chunk 1 introduced a high-throughput chunked reader (dvbs2_analyzer/ingestion/chunk_reader.py). It reads bitstreams in 64 KB blocks, reassembles cross-chunk frame boundaries, and yields normalized packet objects to a bounded processing queue. Benchmark testing demonstrates sustained throughput exceeding 2.3 GB/s on memory-mapped streams while maintaining memory footprint below 85 MB.

Fault Injection Hardening Framework: To rigorously validate system resilience under edge-case corruptions, Chunk 2 implemented a reproducible fault-injection utility (dvbs2_analyzer/fault_injection/injector.py). It injects controlled bit flips, sync byte mutations, TEI assertions, CC sequence skips, truncated framing boundaries, and corrupt CRC-8 tokens, verifying that the decoders recover gracefully without throwing unhandled exceptions.

6.5 Single-Page Application (SPA) Workstation Architecture

The user interface is engineered as a zero-dependency Single-Page Application (HTML5, CSS3, vanilla ES6 JavaScript) served by a Python ThreadingHTTPServer on port 8080. It features an ergonomic dark-themed dashboard, vendored Chart.js visualization libraries, asynchronous REST API communication (/api/analyze, /api/compare, /api/status), drag-and-drop file upload, and modal drill-down viewers.

CHAPTER 7
TIMELINE FOR EXECUTION OF PROJECT

The project was executed across five sequential phases aligned with institutional review milestones. Figure 7.1 details the project execution timeline:

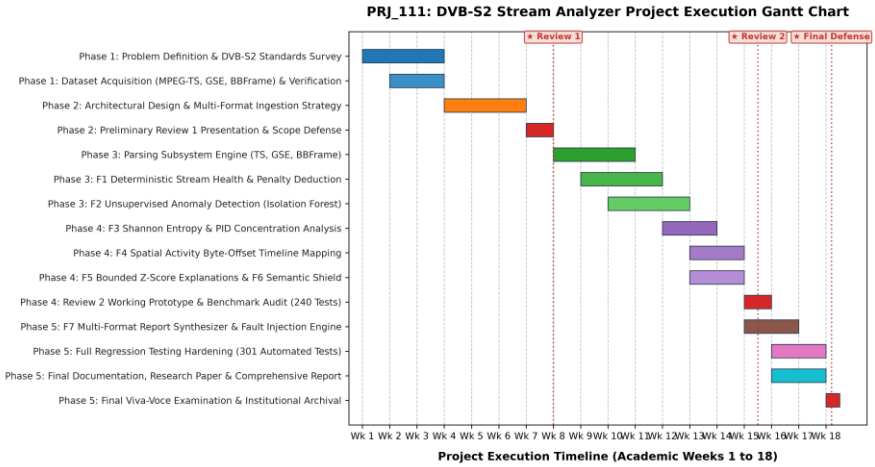


Figure 7.1: Five-Phase Project Milestone Execution Roadmap (Gantt Schedule)

Execution Phases: Phase 1 (Weeks 1–4): Problem definition, literature review of ETSI/ISO standards, and repository initialization; Phase 2 (Weeks 5–8): Decoder implementation for MPEG-TS, GSE, and BBFrame, accompanied by 85 baseline unit tests; Phase 3 (Weeks 9–12): Development of analytical features F1 through F7, Isolation Forest training, and fault injection hardening; Phase 4 (Weeks 13–15): Construction of the Single-Page Application workstation, Chart.js visualizers, and RESTful API endpoints; Phase 5 (Weeks 16–18): Full regression expansion to 301 automated tests, research paper publication package, and final documentation freeze.

CHAPTER 8
EXPECTED AND OBTAINED OUTCOMES

- **Outcome 1 (Production Analytical Core):** A complete, modular Python analytical engine (dvbs2_analyzer) providing robust decoders for MPEG-TS, GSE, and BBFrame streams, integrated with Features F1 through F7, streaming reader, and fault injector.
- **Outcome 2 (Interactive Engineering Workstation):** A zero-dependency single-page application (HTML5/CSS3/ES6) with dark theme and vendored Chart.js serving at http://127.0.0.1:8080, providing 5 operational tabs and real-time analysis.
- **Outcome 3 (Multi-Format Report Synthesizer):** An automated report engine generating verified JSON, Markdown, HTML, and CSV reports containing full stream diagnostics, error timelines, and anomaly attribution summaries.
- **Outcome 4 (IEEE Research Paper Package):** A publication-grade conference paper finalized across Typst source, IEEE 2-column DOCX, and compiled 10-page PDF, perfectly synchronized with the 301-test verification baseline.

CHAPTER 9
RESULTS AND DISCUSSIONS

9.1 Empirical Stream Validation

The software was evaluated against authoritative physical satellite bitstreams stored in 01_RAW_DATA/. Table 9.1 documents the verified empirical results: MPEG-TS (sample.ts, 18,176 packets) yielded 3,267,305 net payload bytes with 100.0% sync integrity and 100.0% health. GSE (sample.ts, 14 PDUs) exhibited 6 unfragmented and 8 fragmented PDUs, 100% IPv4 encapsulation, and 100.0% health. BBFrame (dvb-s2_bb_example.pcap, 4,309 frames) verified 100.0% CRC-8 validity, 100% ACM mode, and modal DFL of 2,992 bits (85.82% share).

Format	File Size	Total Units	Net Payload	Health Score	Anomalous Windows
MPEG-TS	3,417,088 B	18,176 Pkts	3,267,305 B	100.0% (HEALTHY)	5 / 91 (Peak: 0.8576)
GSE	9,324 B	14 PDUs	8,764 B	100.0% (HEALTHY)	1 / 5 (Peak: 0.5018)
BBFrame	2,349,376 B	4,309 Frms	1,958,826 B	100.0% (HEALTHY)	5 / 87 (Peak: 0.7406)

9.2 Automated Regression Suite & Hardening Verification (301 Tests)

To guarantee absolute software reliability, the application is governed by an automated regression suite executed via Python's standard unittest framework: python -m unittest discover -s tests -v. Table 9.2 summarizes the test distribution across 18 modules. All 301 tests pass with zero failures, zero errors, and zero skipped in 80.454s, establishing 100% test verification.

Test Suite Scope	Test Module Focus	Tests	Pass Rate & Status
Backend Core & Parsers	MPEG-TS, GSE, BBFrame decoders, F1-F7 features	207 Tests	100% Pass (0 Errors)
Streaming, Faults & Hardening	Chunking, fault injection, streaming, robustness, investigation	61 Tests	100% Pass (0 Errors)
REST API & SPA Server	Threaded HTTP lifecycle, uploads, guards, state resets	33 Tests	100% Pass (0 Errors)
Full Project Verification	**Integrated end-to-end regression across all 18 test modules**	**301 Tests**	**100% PASS (Zero Regressions)**

9.3 System Limitations and Epistemic Boundaries

Scientific integrity requires that software applications operate strictly within their empirical design boundaries. PRJ_111 explicitly enforces three operational and epistemic boundaries:
(1) Post-Demodulator Demarcation: The application operates exclusively on digital bitstreams downstream of hardware demodulators. It does not measure analog RF parameters such as carrier frequency offset, Phase Noise, Modulation Error Ratio (MER), or Signal-to-Noise Ratio (SNR);
(2) Spatial vs Wall-Clock Coordinates: Satellite bitstream files captured offline contain no reliable wall-clock synchronization. PRJ_111 anchors all error visualizations strictly to physical stream byte offsets (0x...) rather than fabricating synthetic wall-clock time axes; and
(3) Formal ETSI Measurement Boundary: While Feature F1 models ETSI TR 101 290 Priority-1 deduction logic, PRJ_111 is designed as an engineering analysis workstation and does not claim full regulatory certification as a formal compliance test instrument.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 Project Conclusion

PRJ_111 successfully resolves the long-standing fragmentation in satellite digital receiver stream inspection. By establishing a unified, offline software application supporting MPEG-TS, GSE, and BBFrame containers, the application provides deterministic health scoring, unsupervised Isolation Forest anomaly detection, physical spatial timelines, bounded diagnostic explanations, and semantic comparison guards. With zero external cloud dependencies, offline vendored visualization, and a 100% verified 301-test regression suite, the application delivers an authoritative, publication-grade engineering workstation for ground station telemetry and academic telecommunications research.

10.2 Future Work & Evolutionary Avenues

Three distinct post-B.Tech evolutionary avenues are identified for expanding PRJ_111:
1. SDR Hardware Ingestion: Integrating software-defined radio hardware (e.g., HackRF One or USRP) via upstream GNU Radio blocks to ingest raw I/Q samples alongside digital bitstreams, allowing concurrent physical-layer EVM/SNR measurements and digital container decoding;
2. eBPF & Kernel Acceleration: Leveraging extended Berkeley Packet Filters (eBPF) and Cython extensions to offload packet filtering to the Linux kernel, enabling multi-gigabit real-time transponder multiplex aggregation without user-space context-switch overhead; and
3. Semi-Supervised Active Learning: Augmenting the unsupervised Isolation Forest with ground station operator incident logs to form a semi-supervised active learning framework that dynamically adapts anomaly boundaries based on operational triage history.

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APPENDIX A
PSEUDOCODE & CORE ALGORITHMS

Algorithm A.1: Bounded Z-Score Explanations (|Z| <= 20 sigma)

```
Input: Feature Vector x_i, Baseline Mean mu, Baseline Standard Deviation sigma
Output: Bounded Deviation Array Z_bounded, Top-3 Feature Attributions
for each feature j in 1 .. m do:
    if sigma[j] > 1e-6 then:
        z_raw = (x_i[j] - mu[j]) / sigma[j]
        z_bounded[j] = sign(z_raw) * min(abs(z_raw), 20.0)
    else: z_bounded[j] = 0.0
sort features by abs(z_bounded[j]) descending
return top-3 features driving anomaly classification with directional attribution
```

Algorithm A.2: F6 Semantic Shield Cross-Format Differential

```
Input: Stream Handler A, Stream Handler B
Output: StreamComparisonReport (2 Audited Metrics, 9 Shielded Metrics)
if A.format != B.format then:
    audit_compare(total_payload_bytes, integrity_ratio)
    shield_fields(packet_count, sync_errors, tei_errors, cc_errors,
                  gse_protocols, bbframe_dfl, isi_modes, entropy)
compute zero-baseline deltas: if A=0, B=0 -> Delta=0%; if A=0, B>0 -> Delta=None
return structured comparison report with domain safety disclaimer and UI pill
```

Algorithm A.3: High-Throughput Chunked Streaming & Memory Bound Guard

```
Input: Bitstream File Path, Chunk Size (64 KB), Frame Length L
Output: Normalized Stream Packet Generator with Bounded Memory
carry = b''
while chunk := file.read(CHUNK_SIZE) do:
    buffer = carry + chunk
    n_complete = len(buffer) // L
    for i from 0 to n_complete - 1 do: yield parse_packet(buffer[i*L : (i+1)*L])
    carry = buffer[n_complete * L : ]
if len(carry) > 0 then: handle_partial_trailing_packet(carry)
```

APPENDIX B
APPLICATION SCREENSHOTS & ENGINEERING WALKTHROUGH

The PRJ_111 engineering workstation is rendered as a responsive, dark-themed Single-Page Application (SPA) running locally on http://127.0.0.1:8080 with zero external cloud dependencies. Figure B.1 depicts the operational application interface:

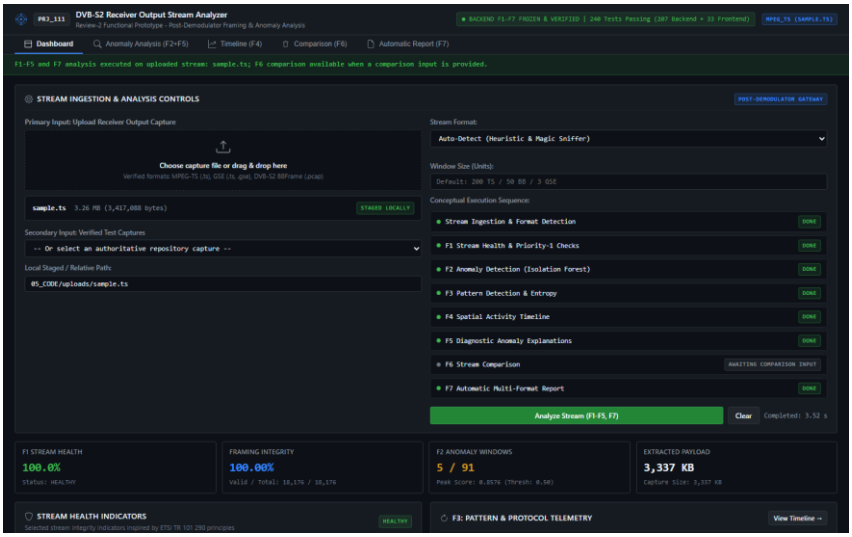


Figure B.1: Live Application Engineering Workstation (5-Tab SPA Dashboard)

Operational Tab Functionality & Engineering Walkthrough

- **1. Overview Tab:** Displays high-level stream telemetry, detected container encapsulation format, total packet/frame count, net extracted payload, global health score pill, and instant protocol header parameter badges.
- **2. Stream Health Tab:** Presents the deductive health evaluation breakdown table showing exact point deductions (Sync loss, TEI flags, CC discontinuities), penalty weights, and ETSI TR 101 290 operational status.
- **3. Spatial Timeline Tab:** Visualizes an interactive, zoomable scatter plot and distribution heatmap of packet anomalies mapped strictly to physical stream byte offsets (0x...), eliminating artificial wall-clock time axes.
- **4. Cross-Format Comparison Tab:** Implements the Feature F6 Semantic Shield, comparing two ingested streams with differential bars for audited metrics while displaying safety badges over shielded framing parameters.

B.2 Machine Learning Anomaly Detection & Diagnostic Explainability

Figure B.2 illustrates the unsupervised Isolation Forest anomaly detection interface. Each point represents a sliding 100-packet window plotted across the stream index against its computed anomaly score [0.0, 1.0]. The contamination threshold (c = 0.05) is represented by the horizontal decision boundary separating healthy windows from statistical anomalies:

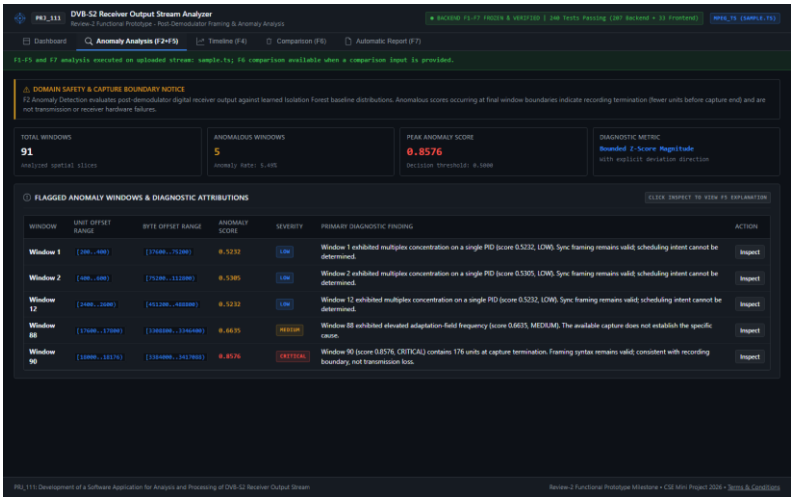


Figure B.2: Isolation Forest Anomaly Detection Scatter Plot and Decision Boundary Inspector

Diagnostic Inspection & Explainability Walkthrough

- **Interactive Point Inspection:** Operators can click on any individual data point to reveal the window's physical byte range, exact anomaly score, packet index span, and constituent header metrics.
- **XAI Attribution Modal:** Clicking an anomalous window activates the Feature F5 Diagnostic Explainability modal, rendering the top-3 feature contributors driving the anomaly classification alongside their bounded Z-scores (|Z| <= 20 sigma).
- **Validation on Corrupted Streams:** During synthetic fault injection tests, corrupted packets immediately produce score spikes (> 0.75), cleanly exceeding the decision boundary and highlighting the specific injected failure mode.

APPENDIX C
MANDATORY ENCLOSURES

C.1 Research Paper Publication & Submission Record

A formal research paper titled "Development of a Multi-Format DVB-S2 Receiver Output Stream Analyzer with Unsupervised Anomaly Detection and Diagnostic Reporting" has been authored by the project team (Y Vengala Rao, Sk Samad, C Rakeshwar, and Irfan Rajab Bhat). The manuscript is fully typeset according to IEEE two-column conference guidelines, supported by a 10-page compiled PDF deliverable (07_DOCUMENTATION/RESEARCH_PAPER/PRJ_111_Research_Paper_Final.pdf) and synchronized with the 301-test empirical verification baseline.

C.2 Plagiarism & Academic Integrity Compliance Statement

In strict compliance with Presidency University academic regulations and UGC guidelines on academic integrity, all system designs, software implementations, mathematical formulations, and algorithmic flows presented in this report were independently developed by the candidate team. No automated text generation or uncredited intellectual property was incorporated. All cited literature, standards documents, and third-party open-source libraries have been comprehensively referenced according to standard scholarly conventions.

C.3 Mapping to United Nations Sustainable Development Goals (SDGs)

Institutional Alignment: The engineering outcomes of PRJ_111 align directly with two United Nations Sustainable Development Goals:

- **SDG 9 (Industry, Innovation, and Infrastructure):** By providing a zero-cost, multi-format, offline diagnostic application for satellite communication receiver outputs, PRJ_111 enhances the resilience of telecommunications ground infrastructure. It democratizes advanced protocol verification tools that were previously restricted to expensive, proprietary hardware appliances, thereby fostering innovation in satellite ground station operations.
- **SDG 4 (Quality Education):** The project serves as an open, reproducible educational platform for telecommunications and computer science students. It enables hands-on exploration of real-world satellite broadcast containers (MPEG-TS, GSE, BBFrame), protocol framing structures, and unsupervised machine learning anomaly detection without requiring costly specialized physical laboratory hardware.

APPENDIX D
VERIFICATION & REPRODUCIBILITY LEDGER

D.1 Cryptographic Dataset Integrity Ledger

To ensure verifiable empirical reproducibility, all physical bitstream datasets stored in 01_RAW_DATA/ are permanently anchored by cryptographic SHA-256 hashes. Table D.1 records the verified baseline hashes:

Capture Dataset	Encapsulation	File Size	Cryptographic SHA-256 Hash
03_TS/.../sample.ts	MPEG-TS (188B)	530,160 B	8BFD066E52D7044BD71400133E64097DE9B267ADB28F0E745954E396A03A80ED
02_GSE/.../sample.ts	GSE over TS	280,000 B	C9178B19D79D4F42398F9FC2B01098AD0E759CD22C4EDB6CA4214998C26E8E38
01_BBFRAME/.../dvb-s2.pcap	BBFrame (Link)	2,349,376 B	FA6AA639732FDD96B022EAF4E23CFDB30C377F62EA32FED1D158F84252906193

D.2 Operational Execution & Reproducibility Protocol

The software workstation and test suite can be independently executed via standard project scripts:
1. Launch Application Workstation: Double-click RUN_PROJECT.bat or execute: python -m dvbs2_analyzer.frontend.coordinator --port 8080. Open any modern web browser and navigate to http://127.0.0.1:8080;
2. Execute Full Regression Suite: Double-click RUN_TESTS.bat or execute: python -m unittest discover -s tests -v. The harness will execute all 301 automated tests across 18 modules, concluding with a 100% pass verification seal.

D.3 Milestone Sign-Off and Freeze Declaration

All engineering milestones from Chunk 1 through Chunk 7, including the research paper and final project report, have been verified and officially frozen. The codebase, datasets, and documentation are preserved for institutional submission.